

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Date of Assignment : 05-08-2026	Date/Time of Submission : 05-08-2026

Hash Table:

Write a program to:

- Implement a hash table with separate chaining for collision handling.

Code:

```
Practical 7A.py - C:/Users/itlab/Desktop/SYIT S039/Data strutrue Practical/Practical 7A.py (3.14.6)
File Edit Format Run Options Window Help

size = 5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"Abhishek.Y")
insert(15,"Pradeep.R")
insert(18,"Divyanshu.S")
insert(20,"Abhishek.P")
insert(30,"Dhruv.G")

print(table)
print("\nDivyanshu Singh S039")
```

Output:

```
= RESTART: C:/Users/itlab/Desktop/SYIT S039/Data strutrue Practical/Practical 7A.py
[[ (10, 'Abhishek.Y'), (15, 'Pradeep.R'), (20, 'Abhishek.P'), (30, 'Dhruv.G') ], [], [], [(18, 'Divyanshu.S')], [] ]
Divyanshu Singh S039
```

- Store and retrieve data from the hash table.

Code:

Practical 7A.py - C:/Users/itlab/Desktop/SYIT S039/Data strutrue Practical/Practical 7A.py (3.14.6)

File Edit Format Run Options Window Help

```
size = 5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not found"

insert(10,"Abhishek.Y")
insert(15,"Pradeep.R")
insert(18,"Divyanshu.S")
insert(20,"Abhishek.P")
insert(30,"Dhruv.G")

print(table)

key = int(input("Enter Key : "))
print(search(key))

print("\nDivyanshu Singh S039")
```

Output:

```
===== RESTART: C:/Users/itlab/Desktop/SYIT S039/Data strutrue Practical/Practical 7A.py =====
[[ (10, 'Abhishek.Y'), (15, 'Pradeep.R'), (20, 'Abhishek.P'), (30, 'Dhruv.G') ], [], [], [(18, 'Divyanshu.S')], []]
Enter Key : 10
Abhishek.Y
Divyanshu Singh S039
```